DIY-AI

Running Large Language and Agentic Models in your own environments

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Local AI: models, runtimes and hardware

A trained **model** supplies the weights. A **runtime** loads them and runs inference on compatible **hardware**.

Runtime	Example model	Workshop use
Ollama service	`gemma4:26b`	Text chat on Mac
vLLM Metal service	`mlx-community/gemma-4-e4b-it-4bit`	Text, no audio in this setup
vLLM Metal service	`openai/whisper-large-v3`	Audio transcription
MLX-VLM program	`google/gemma-4-E4B-it`	Audio input on Mac
vLLM service	`deepseek-ai/DeepSeek-V4-Flash`	Text and code on Spark

Local AI is practical: match your task and model to the runtime and available memory.

Workshop configurations from the outline, notebooks and launch scripts. Runtime references: vLLM Metal, MLX-VLM, Ollama.

Four ways to run the workshop models

Hardware and runtime	How we use it	Entry point
Apple Silicon + Ollama	Easiest starting point for local chat	`ollama run gemma4:26b`
Apple Silicon + vLLM Metal	A local model service for notebooks and apps	`./scripts/start-mlx-vllm.sh` API at `localhost:8000`
Apple Silicon + MLX-VLM	Direct Python inference, including Gemma audio	`notebooks/local-vlm-gemma.ipynb`
DGX Spark + vLLM	Connect over VPN and call the remote model service	Change the API base URL to Spark

After installation and download, start a model with a command or a cell.
Gemma's saved local model load: **4.8 seconds**.

Sources: local launch scripts and notebooks. Spark audio uses 1 DGX Spark. DeepSeek uses 2 DGX Sparks. Hardware and VPN configuration follow the workshop outline.

Three tests of useful model behavior

1. Five questions: observed deployment performance

Capital of France, an AI poem, quantum computing, the laws of robotics and the plot of *Inception*.

Measures: request latency, completion tokens per second and cut-off answers.

2. Dutch transcription: agreement with a curated reference

Jeroen Hendrikse's radiology recording from UMC Utrecht.

Measure: word error rate (WER) over 1,350 reference words.

3. LiveCodeBench-Pro: executable programming solutions

Ten contest problems, with one C++ solution attempt per problem.

Measure: accepted solutions out of ten selected problems.

Only DeepSeek on Spark produced judgeable code among our self-hosted setups.

GPT-6 Astra provides a separate cloud comparison.

Sources: original presentation, transcription notebooks and saved LiveCodeBench-Pro results. The five-question smoke test describes deployments, without a controlled speed or correctness comparison.

Model size is not the same as inference cost

Dense small model → sparse MoE → very large sparse MoE

Model	Total parameters	Active/effective per token
Gemma 4 E4B (dense)	∼8B incl. embeddings	∼4.5B effective
Gemma 4 26B A4B (MoE)	∼25.2B	∼3.8–4B active
DeepSeek V4 Flash (MoE)	∼284B	∼13B active

Total describes stored model capacity. **Active/effective** better indicates per-token computation, but does not determine tokens/s alone.

Gemma 26B can be faster than E4B: mixture of experts (MoE) selects only ∼4B parameters per token. The full 26B is not evaluated each time.

DeepSeek: ∼284B parameters of model capacity, ∼13B active per token.

E4B is dense, not MoE. Its large embedding tables use lookups, explaining the effective count. Parameter counts are approximate, not FLOP or speed measurements. Sources: Google Gemma 4 model card; DeepSeek V4 release.

Observed inference performance: three local deployments

Request latency	**vLLM Metal** Gemma 4 E4B	**Ollama** Gemma 4 26B A4B	**Distributed vLLM** DeepSeek V4 Flash
Capital of France	1.745 s	1.478 s	0.607 s
AI poem	5.491 s	19.619 s	3.686 s
Quantum computing	7.818 s	22.523 s	23.756 s
Laws of robotics	7.463 s	14.958 s	8.349 s
Inception	7.794 s	24.565 s	15.920 s
Mean latency	6.062 s	16.628 s	10.464 s
Completion rate (tokens/s)	**18.64**	**34.37**	**28.94**

Real-world local deployment trade-offs, not a controlled model-speed benchmark.

Original 7 October 2026 snapshot. Completion rate = total reported completion tokens / total request time (including prompt processing and request overhead). Hardware, serving stacks, formats/quantization, tokenizers and reasoning behavior differ.

Why these measurements are not apples-to-apples

Model	**Workshop deployment**
Gemma 4 E4B	4-bit MLX model, vLLM Metal on M1 Pro
Gemma 4 26B A4B	Ollama on M1 Pro
DeepSeek V4 Flash	Distributed vLLM across 2× DGX Spark

- **Serving stacks, hardware, formats and quantization differ.**
- **Tokenizers and reasoning differ,** so token counts are not equivalent.

Dutch transcription: three models on one recording

Jeroen Hendrikse, UMC Utrecht: *radiologieverslag*
473.6 seconds of speech, scored against 1,350 curated reference words.

Model	**Runtime / hardware**	**WER ↓**	**Time**
Whisper large-v3 `openai/whisper-large-v3`	vLLM Metal Apple M1 Pro	11.93%	53.4 s
Gemma 4 E4B `google/gemma-4-E4B-it`	MLX-VLM Apple Silicon	**7.70%**	144.3 s
Qwen3-Omni 30B A3B `Qwen/Qwen3-Omni-30B-A3B-Instruct`	vLLM 1 DGX Spark	8.52%	98.1 s

Gemma has the lowest WER. Whisper has the shortest recorded time.

Sources: saved transcription notebooks, re-scored 23 October 2026; Whisper timing: Apple Silicon validation, 23 October 2026. Whisper: full transcription request cell. Gemma/Qwen: chunk loops, excluding model loading and initial audio decoding. Different runtimes and timing scopes.

Word error rate: Gemma's small advantage

$$\mathrm{WER} = \frac{\text{substitutions} + \text{deletions} + \text{insertions}}{\text{reference words}} \times 100\%$$

Model	Substitutions	Deletions	Insertions	Total
Whisper large-v3	85	41	35	161
Gemma 4 E4B	68	21	15	**104**
Qwen3-Omni	74	15	26	115

Gemma: 11 fewer word errors than Qwen

7.70% versus 8.52% WER: a **0.81 percentage-point** gap using the unrounded scores.

Scoring ignores case and punctuation, splits hyphens, and retains fillers and repetitions. Curated TXT and SRT references yield identical scores.

Sources: the three transcription notebooks, cell index 7. One recording and different inference pipelines. This small difference does not establish a general model ranking. WER counts word differences without weighting their clinical meaning.

Gemma 4: evaluation of a five-sentence summary

Gemma 4 summarized the Dutch transcription in **five English sentences**, then evaluated the summary against the full input.

Gemma's ratings (1–5)

Criterion	Score
Factual faithfulness	5/5
Main-point coverage	5/5
Preserving uncertainty	4/5
English clarity	5/5

Checks against the input

- **Evaluator error:** reported 4 sentences; the summary actually has **5**.
- **Missing detail:** the ischemic finding was *new since the previous MRI*.
- **Precision:** "residual tumor" is clearer; the four-point check is compressed imprecisely.

Broadly faithful summary; the evaluator also needs checking.

26 October 2026. Summary and evaluation: `mlx-community/gemma-4-e4b-it-4bit`, local vLLM. Scores are subjective self-evaluation, not an independent benchmark. Source: saved summary notebook and local evaluation report.

LiveCodeBench-Pro: a harder programming test

Codeforces, ICPC and IOI problems require algorithms and code that pass the judge's tests.

What happened locally

Local MLX trials yielded no usable solutions.
The saved judged comparison records:

- **Ollama, Gemma 4 26B A4B:** all 10 responses hit the 4,096-token limit without final code.
- **Local vLLM: Not run.**

The larger sparse MoE model

DeepSeek V4 Flash

`deepseek-ai/DeepSeek-V4-Flash`

$\sim$284B total, $\sim$13B active per token

vLLM over VPN on **2 DGX Sparks**
256 GB total unified memory
(128 GB per node)

One C++ attempt per problem.
Temperature 0, 4,096 tokens.

Task difficulty and output budgets limit these trials. They do not establish a general model limit.

Local MLX observation and cluster configuration: workshop outline. Saved evidence: Ollama, 8 October; Spark, 20 October 2026. LiveCodeBench-Pro paper; DGX Spark specifications.

LiveCodeBench-Pro: Spark and GPT-6 Astra

Same ten problems and ARM64 judge. Both runs completed judging for all ten.

	DeepSeek V4 Flash 2 DGX Sparks, 20 October	**GPT-6 Astra** Codex session, 21 October
Accepted / selected	**4/10 (40%)**	**10/10 (100%)**
Compilation error	4	0
Wrong answer	2	0
Mean inference time	25.32 s	Not measured

Astra passed all ten problems. DeepSeek's six failures included four compilation errors.

Generation conditions differ:

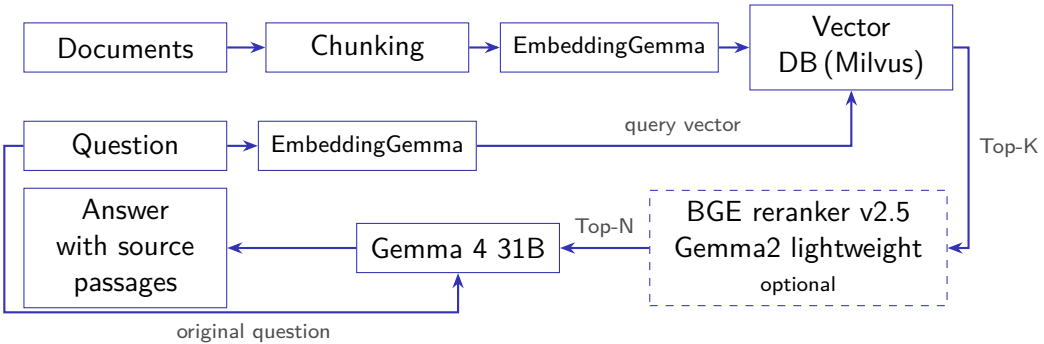
DeepSeek: temperature 0, 4,096-token API budget per problem.

Astra: Codex with `xhigh` reasoning, shared conversation context, no per-problem token cap.

Source: livecodebench-gpt-6-astra-results.tex and saved JSON. One judged submission per problem. Astra sources frozen before testing, with no revisions after feedback. Subset: 7 easy, 3 medium, 0 hard. Generation conditions differ. These are workshop results, not full-benchmark leaderboard scores.

Local document search with RAG on DGX Spark

Retrieval-augmented generation (RAG) supplies relevant document passages to the model before it writes an answer.



Top-K retrieves candidate passages. The optional reranker selects Top-N. Without reranking, the retrieved passages go directly to Gemma.

Proposed local stack from the outline, not a benchmarked deployment. Model references: EmbeddingGemma 300M, BGE reranker, Gemma 4 31B; Milvus documentation.

Embeddings: meaning as coordinates

Text becomes a vector

An embedding is a **list of numbers** representing text.

Similar meanings tend to lie nearby in a high-dimensional space.

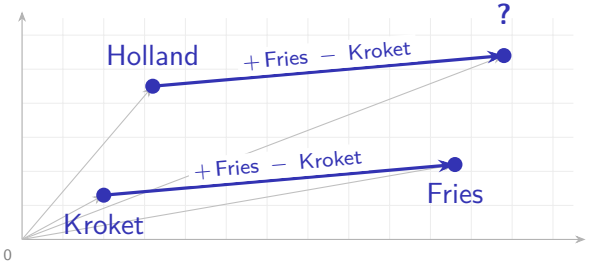
Meaning has direction

$$E(\text{King}) + E(\text{Woman}) - E(\text{Man}) \approx E(\text{Queen})$$

$E(\text{word})$ is its embedding vector.

Quiz: which country?

Holland + Fries − Kroket ≈ **?**



Illustrative 2D embedding space; not a measured model result.

Back to RAG: EmbeddingGemma represents questions and document chunks in a shared space so similarity search can retrieve relevant passages.

Embedding intuition and word analogy: 3Blue1Brown, Transformers, the tech behind LLMs. Text retrieval: EmbeddingGemma model card.

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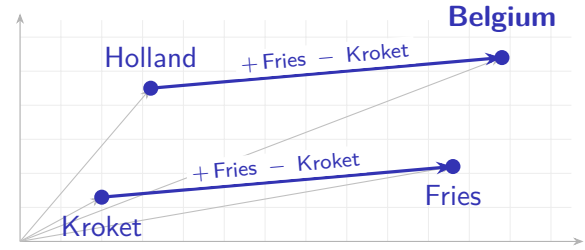
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Quiz: which country?

Holland + Fries − Kroket ≈ **Belgium**



Illustrative 2D embedding space; not a measured model result.

Add the fries association and subtract the kroket association.

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Next: agentic orchestration in clinical imaging

Local models and document retrieval are building blocks for a wider workflow.

The next presenter will introduce an agentic orchestrator for use in clinical imaging practice.

Coordinating model calls, retrieved information and tools around the steps of a clinical imaging workflow.

Over to the next presenter.

Try it after the presentation



Scan to download the workshop